

AWS Project

Using AWS CodeCommit, CodePipeline, CodeBuild, CodeDeploy, EC2, ALB, ASG, S3, IAM, and CloudWatch, design a complete CI/CD pipeline to deploy an application with high availability.

AWS CI/CD PIPELINE — FULL DETAILED IMPLEMENTATION GUIDE

● STEP 0: Prerequisites (Before Starting)

Make sure you have:

- ✓ AWS Account
- ✓ IAM User with Admin Access
- ✓ Git Installed
- ✓ AWS CLI Installed & Configured

Check:

```
aws --version
```

```
aws configure
```

● STEP 1: Create IAM Roles (Security Foundation)

📌 AWS Console → IAM → Roles → Create role

1 CodePipeline Role

Service:

CodePipeline

Policy:

AdministratorAccess

Name:

CodePipelineRole

2 CodeBuild Role

Service:

CodeBuild

Policy:AdministratorAccess

Name:CodeBuildRole

3 CodeDeploy Role

Service:CodeDeploy

Policy:AWSCodeDeployRole

Name:CodeDeployRole

4 EC2 Role

Service:EC2

Attach Policies:

AmazonS3ReadOnlyAccess

CloudWatchAgentServerPolicy

AmazonEC2RoleforAWSCodeDeploy

Name:EC2CICDRole

👉 Purpose: Allows EC2 to talk to S3, CodeDeploy, CloudWatch.

● STEP 2: Create CodeCommit Repository

📌 AWS → CodeCommit → Create Repository

Name:cicd-webapp

Clone to Local

```
git clone <repo-url>
```

```
cd cicd-webapp
```

Create Project Files

Inside folder:

```
touch index.html buildspec.yml appspec.yml
```

```
mkdir scripts
```

```
touch scripts/start.sh
```

Sample index.html

```
<h1>Hello AWS CI/CD</h1>
```

Push Code

```
git add .
```

```
git commit -m "Initial commit"
```

```
git push origin main
```

● STEP 3: Create S3 Bucket (Artifact Storage)

📌 AWS → S3 → Create bucket

Name:cicd-artifacts-2026

Region: Same region

Settings:

✓ Block Public Access → ON

✓ Versioning → Optional

Create bucket.

👉 Stores build ZIP files.

● STEP 4: Create Security Group

📌 EC2 → Security Groups → Create

Name:cicd-sg

Inbound Rules:

Type	Port	Source
SSH	22	My IP
HTTP	80	0.0.0.0/0

Save.

● STEP 5: Create Launch Template

📌 EC2 → Launch Templates → Create

Basic Config

Name:cicd-template

AMI:Amazon Linux 2

Instance:t2.micro / t3.micro

Key Pair:your-key

Security Group:cicd-sg

IAM Role:

EC2CICDRole

User Data (VERY IMPORTANT)

Paste:

```
#!/bin/bash
```

```
yum update -y
```

```
# Install Apache
```

```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
# Install CodeDeploy Agent
```

```
yum install ruby wget -y
```

```
cd /home/ec2-user
```

```
wget https://aws-codedeploy-ap-southeast-2.s3.ap-southeast-2.amazonaws.com/latest/install
```

```
chmod +x install
```

```
./install auto
```

```
systemctl start codedeploy-agent
```

Create Template.

👉 Makes every EC2 auto-ready.

● STEP 6: Create Auto Scaling Group

📌 EC2 → Auto Scaling Groups → Create

Select:

Launch Template → cicd-template

VPC → Default

Subnets → All

Capacity:

Min: 2

Desired: 2

Max: 3

Health Check:

EC2 + ELB

Create.

👉 Keeps 2 servers always.

● STEP 7: Create Load Balancer

📌 EC2 → Load Balancers → Create

Type:

Application Load Balancer

Basic

Name:

cicd-alb

Scheme:

Internet-facing

Target Group

Type:

Instance

Port:

80

Health Check:

/

Register ASG.

Create ALB.

👉 Routes traffic.

● STEP 8: Configure CodeDeploy

📌 AWS → CodeDeploy → Create Application

Application Name:

cicd-app

Compute:

EC2/On-Premises

Deployment Group

Name:

cicd-dg

Service Role:

CodeDeployRole

Environment:

Auto Scaling Group

Load Balancer:

✓ Enable

Select:

ALB

Target Group

Save.

● STEP 9: Configure buildspec.yml

Inside repo:

version: 0.2

phases:

install:

commands:

- echo Installing...

build:

commands:

- echo Building...

artifacts:

files:

- '**/*'

Commit & Push.

● STEP 10: Configure appspec.yml

version: 0.0

os: linux

files:

- source: /

destination: /var/www/html

hooks:

AfterInstall:

- location: scripts/start.sh

timeout: 300

● STEP 11: Configure start.sh

scripts/start.sh

```
#!/bin/bash
```

```
systemctl restart httpd
```

Make executable:

```
chmod +x scripts/start.sh
```

Push again.

● STEP 12: Create CodeBuild Project

📌 AWS → CodeBuild → Create Project

Name:

cicd-build

Source:

CodeCommit

Repo:

cicd-webapp

Environment:

Amazon Linux

Standard Image

Service Role:

CodeBuildRole

Artifacts:

S3

Bucket: cicd-artifacts-2026

Save.

● STEP 13: Create CodePipeline

📌 AWS → CodePipeline → Create

Name:

cicd-pipeline

Role:

CodePipelineRole

Stage 1: Source

Provider: CodeCommit

Repo: cicd-webapp

Branch: main

Stage 2: Build

Provider: CodeBuild

Project: cicd-build

Stage 3: Deploy

Provider: CodeDeploy

App: cicd-app

Group: cicd-dg

Create.

 Pipeline is live.

● STEP 14: Configure CloudWatch (Monitoring)

Login EC2:

```
sudo yum install amazon-cloudwatch-agent -y
```

Wizard:

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
```

Start:

```
sudo systemctl start amazon-cloudwatch-agent
```

Check:

```
sudo systemctl status amazon-cloudwatch-agent
```

● STEP 15: Test End-to-End

Edit index.html:

```
vi index.html
```

Change text.

Push:

```
git commit -am "Update app"
```

```
git push
```

Pipeline runs automatically.

Check:

✓ CodePipeline → Success

✓ CodeDeploy → Success

✓ ALB → Healthy

✓ Browser → App Updated

 STEP 16: Access Application

 EC2 → Load Balancer → DNS

Example:

<http://cicd-alb-123456.elb.amazonaws.com>

Open in browser.

COMPLETE WORKFLOW

Developer



CodeCommit



CodePipeline



CodeBuild → S3



CodeDeploy



EC2 (ASG)



ALB



Users



CloudWatch

PERFECT INTERVIEW ANSWER

“I configured IAM roles, created CodeCommit, CodeBuild, CodeDeploy, and CodePipeline.

The app is deployed on EC2 using ASG and ALB.

CloudWatch is used for monitoring.

The entire process is automated from code push to production.”

Search

[Alt+S]

Global

Darsani (3695-0434-1981)

EC2-CodeDeploy-Role

Info

Delete

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Access reports

Summary

Creation date
January 28, 2026, 11:03 (UTC+05:30)

ARN
arn:aws:iam::369504341981:role/EC2-CodeDeploy-Role

Instance profile ARN
arn:aws:iam::369504341981:instance-profile/EC2-CodeDeploy-Role

Last activity
10 minutes ago

Maximum session duration
1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (3)

Info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type
All types

Policy name

Type

Attached entities

☐

AmazonS3ReadOnlyAccess

AWS managed

1

CloudShell

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Search

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Global

Darsani (3695-0434-1981)

CodeBuild-Service-Role

Info

Delete

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Access reports

Summary

Creation date
January 28, 2026, 11:05 (UTC+05:30)

ARN
arn:aws:iam::369504341981:role/CodeBuild-Service-Role

Maximum session duration
1 hour

Last activity
33 minutes ago

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (4)

Info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type
All types

Policy name

Type

Attached entities

☐

AmazonS3FullAccess

AWS managed

2

☐

AWSCodeBuildDeveloperAccess

AWS managed

2

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The screenshot shows the AWS IAM console interface. On the left is a navigation sidebar with options like 'Dashboard', 'Access Management', 'Users', 'Roles', 'Policies', etc. The main content area is titled 'CodePipeline-Service-Role'. It includes a 'Summary' section with details like 'Creation date' (January 28, 2026) and 'ARN' (arn:aws:iam:369504341981:role/CodePipeline-Service-Role). Below this is a 'Permissions' section showing a list of 'Permissions policies (7)' attached to the role. The table lists policies like 'AmazonS3FullAccess', 'AWSCodeBuildDeveloperAccess', and 'AWSCodeCommitFullAccess' with their respective types and attached entities.

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Launch Templates (1)

Search

☐	Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
☐	lt-0f0573c2a1828e141	lt-cicd-app	1	1	2026-01-28T06:24:00.000Z	arn:aws:iam::369504

Select a launch template

EC2 > Auto Scaling groups

Auto Scaling groups (1)

Last updated less than a minute ago

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Search your Auto Scaling groups

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max
☐	asg-cicd-app	lt-cicd-app Version Latest	2	-	2	2	4

0 Auto Scaling groups selected

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aws

Search

[Alt+S]

United States (N. Virginia)

Darsani (3695-0434-1981)

Darsani

EC2 > Security Groups

Security Groups (3) Info

Find security groups by attribute or tag

☐

Name

☐

Security group ID

☐

Security group name

☐

VPC ID

☐

Description

☐

-

sg-0c97e15465f649b6b

launch-wizard-1

ypc-06e1b62fd13d83a23

launch-wizard-1 cre

☐

-

sg-02242a5661f4b0494

All TCP

ypc-06e1b62fd13d83a23

Security group is cre

☐

-

sg-0a2159f45733f36e9

default

ypc-06e1b62fd13d83a23

default VPC security

Select a security group

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroups

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ENG IN 02:08 PM 28-01-2026

aws

Search

[Alt+S]

United States (N. Virginia)

Darsani (3695-0434-1981)

Darsani

EC2 > Load balancers

Load balancers (1) What's new?

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

☐

Name

☐

State

☐

Type

☐

Scheme

☐

IP address type

☐

VPC ID

☐

Availability Zones

☐

alb-cicd-app

Active

application

Internet-facing

IPv4

ypc-06e1b62fd13d83a23

6 Availability Zones

0 load balancers selected

Select a load balancer above.

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Developer Tools

CodeDeploy

Source • CodeCommit

Artifacts • CodeArtifact

Build • CodeBuild

Deploy • CodeDeploy

Getting started

Deployments

Applications

Deployment configurations

On-premises instances

Pipeline • CodePipeline

Settings

Go to resource

Feedback

Developer Tools > CodeDeploy > Deployments

Deployment history

View details

Actions

Copy deployment

Retry deployment

Q

< 1 >

	Deployment Id	Status	Deploym...	Compute ...	Application	Deploym...	Revision L...	Initiating ...	Start
	d-2CVIRDIH	In progress	In-place	EC2/On-p...	codedepl...	dg-cicd-app	s3://code...	Autoscali...	Jan 21
	d-B6XWZHIH	In progress	In-place	EC2/On-p...	codedepl...	dg-cicd-app	s3://code...	Autoscali...	Jan 21
	d-WWASDLIH	Succeeded	In-place	EC2/On-p...	codedepl...	dg-cicd-app	s3://code...	User action	Jan 21

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Developer Tools

CodeBuild

Source • CodeCommit

Artifacts • CodeArtifact

Build • CodeBuild

Getting started

Build projects

Build history

Report groups

Report history

Compute fleets **New**

Account metrics

Related integrations

Jenkins

GitHub Actions **New**

GitLab runners **New**

Deploy • CodeDeploy

Go to resource

Feedback

Developer Tools > CodeBuild > Build projects

Build projects info

Actions

Create trigger

View details

Debug build

Start build

Create project

Q

Your projects

< 1 >

	Name	Source provider	Repository	Latest build status	Description	Last Modified
	codebuild-cicd-app	AWS CodeCommit	sample-3tier-app	Succeeded	-	2 hours ago

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File Edit Selection View Go Run Terminal Help

sample-3tier-app

EXPLORER

SAMPLE-3TIER-APP

scripts

install.sh

start.sh

appspec.yml

buildspec.yml

index.html

buildspec.yml

1 version: 0.2

2

3 phases:

4 build:

5 commands:

6 - echo "Build started"

7 - echo "Nothing to compile for this app"

8

9 artifacts:

10 files:

11 - '**/*'

12

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS

Darsani T S@DESKTOP-QEENCIM MINGW64 ~/Desktop/sample-3tier-app (master)

\$ git add .

\$ git add buildspec.yml

git commit -m "Add buildspec for codeBuild"

git push

[master e933c41] Add buildspec for codeBuild

1 file changed, 11 insertions(+)

create mode 100644 buildspec.yml

Enumerating objects: 4, done.

Counting objects: 100% (4/4), done.

Delta compression using up to 12 threads

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 463 bytes | 463.00 KiB/s, done.

Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Validating objects: 100%

To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/sample-3tier-app

3ae286d..e933c41 master -> master

Darsani T S@DESKTOP-QEENCIM MINGW64 ~/Desktop/sample-3tier-app (master)

\$

Developer Tools

CodePipeline

Pipelines

Source • CodeCommit

Artifacts • CodeArtifact

Build • CodeBuild

Deploy • CodeDeploy

Pipeline • CodePipeline

Getting started

Pipelines

Account metrics

Settings

Go to resource

Feedback

Pipelines Info

View history Release change Delete pipeline Create pipeline

1

Name	Latest execution status	Latest source revisions	Latest execution started	Most recent executions
pipeline-cicd-app	Succeeded	Source - e933c416: Add buildspec for CodeBuild	57 minutes ago	View details

aws [Search] [Alt+S] United States (N. Virginia) Darsani (3695-0434-1981) Darsani

EC2 > Target groups

Target groups (1) Info | What's new?

Filter target groups

☐	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☐	tg-cicd-app	arn:aws:elasticloadbalancing...	80	HTTP	Instance	alb-cicd-app	vpc-06e1bf...

0 target groups selected

Select a target group above.

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aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Darsani (3695-0434-1981) Darsani

Developer Tools > CodePipeline > Pipelines > pipeline-cicd-app

pipeline-cicd-app

Latest retry trigger ManualStageRetry

Auto retry attempt 1

Manual retry attempt 4

Source

3e0eab9e-61bb-46c8-aa00-0b69ba044d3e

All actions succeeded.

Source

AWS CodeCommit

16 minutes ago

e933c416 Source: Add builds

Build

3e0eab9e-61bb-46c8-aa00-0b69ba044d3e

All actions succeeded.

Build

AWS CodeBuild

12 minutes ago

e933c416 Source: Add builds

Deploy

3e0eab9e-61bb-46c8-aa00-0b69ba044d3e

All actions succeeded.

Deploy

AWS CodeDeploy

2 minutes ago

e933c416 Source: Add builds

Click on any icon to automatically scroll to the corresponding pipeline stage.

Horizontal Vertical

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CloudWatch

Log management

Log groups

Data sources - new

Summary

Log groups (2)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search

Exact match

Log group

Log class

Anomaly d...

Deletion...

Data ...

Sensit...

Retenti...

Met

/aws/codebuild/codebuild-cicd-app

Standard

Configure

Off

-

-

Never exp...

-

/ec2/httpd/error

Standard

Configure

Off

-

-

Never exp...

-

CloudShell

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CloudWatch

Log management

/ec2/httpd/error

i-0bc4da1767a823101

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events.

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

Timestamp

Message

No older events at this moment.

Retry

2026-01-28T09:14:07.073Z

[Wed Jan 28 08:58:16.668132 2026] [suexec:notice] [pid 4118:tid 4118] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)

2026-01-28T09:14:07.073Z

[Wed Jan 28 08:58:16.689044 2026] [lbmethod_heartbeat:notice] [pid 4118:tid 4118] AH02282: No slotem from mod_heartbeat

2026-01-28T09:14:07.073Z

[Wed Jan 28 08:58:16.689792 2026] [systemd:notice] [pid 4118:tid 4118] AH0497: SELinux is enabled; httpd running as context system_u...

2026-01-28T09:14:07.073Z

[Wed Jan 28 08:58:16.703221 2026] [mpm_event:notice] [pid 4118:tid 4118] AH00489: Apache/2.4.66 (Amazon Linux) configured -- resuming...

2026-01-28T09:14:12.352Z

[Wed Jan 28 08:58:16.703242 2026] [core:notice] [pid 4118:tid 4118] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'

No newer events at this moment.

Auto retry paused.

Resume

CloudShell

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CI/CD Pipeline Working 🚀

Deployed via AWS CodePipeline

```
ec2-user@ip-172-31-72-51:~$ -cloudwatch-agent.json]I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config
/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml
Configuration validation second phase succeeded
Configuration validation succeeded
[ec2-user@ip-172-31-72-51 ~]$ sudo systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled;
   Active: active (running) since Wed 2026-01-28 09:14:05 UTC; 11s ago
   Main PID: 27243 (amazon-cloudwat
   Tasks: 5 (limit: 1120)
   Memory: 53.0M
   CPU: 274ms
   CGroup: /system.slice/amazon-cloudwatch-agent.service
           └─27243 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-age

Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2724
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2725
Jan 28 09:14:05 ip-172-31-72-51.ec2.internal start-amazon-cloudwatch-agent[2724
lines 1-20/20 (END)
[ec2-user@ip-172-31-72-51 ~]$ history
 1 sudo yum install -y amazon-cloudwatch-agent
 2 sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-
wizard
 3 sudo find /opt/aws/amazon-cloudwatch-agent -name "*.json"
 4 sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a
fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s
 5 sudo vi /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json
 6 sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a
fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwat
ch-agent.json -s
 7 sudo systemctl status amazon-cloudwatch-agent
 8 history
[ec2-user@ip-172-31-72-51 ~]$ ^C
[ec2-user@ip-172-31-72-51 ~]$
```